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WHITE PAPER | VERSION 1.1

Institutional Reasoning Engines:

A Modular Architecture for AI-Powered Regulated Investigation

8 Universal Base Components · 7 Modular Clusters · 6 Deployment Profiles

From Interview-Only Investigations to Full Financial Crime Platforms

Applicable to Any Regulated Investigation in Any Industry or Jurisdiction

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Executive Summary

PROBLEM	CATEGORY	THIS PAPER
Regulated investigations require defensible, auditable outputs. Existing AI tools are either too generic to encode investigation methodology or too industry-specific to deploy across contexts.	Institutional Reasoning Engines: AI systems that maintain a hypothesis, iterate toward a defensible conclusion, gate every output through human judgment, and produce a tamper-evident audited record. A new class — not an assistant, not automation.	A modular architecture: 8 universal base components mandatory for any compliant investigation, and 7 plug-and-play clusters activating by evidence type, complexity, and institutional maturity. Deployable from a 2-cluster interview-only setup to a full financial crime platform.

Every regulated investigation faces the same requirement: findings traceable to evidence, human accountability at every material decision, and an auditable reasoning process. This architecture encodes those requirements as engineering constraints — not policies that humans might follow inconsistently. The architecture is jurisdiction-agnostic and industry-agnostic. Financial crime serves as the most fully elaborated deployment example because it presents the most demanding combination of evidence types, regulatory obligations, and evidentiary standards. The underlying architecture applies wherever professional reasoning under uncertainty must produce a defensible, auditable output.

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How to Read This Paper

This paper serves three distinct reader types. Each has a different entry point and a different primary interest. You do not need to read sequentially.

Reader	Primary Interest	Start Here
Decision Maker (Governance head, CISO, CFO, Practice Lead)	Which deployment profile fits your organisation, what it costs, and what it delivers. Whether the architecture is credible enough to invest in.	Sections 2, 11, 14 — Modular Framework, Deployment Profiles, ROI
Implementation Lead (Head of Investigations, Forensic Lead, Compliance Director)	What the mandatory base requires, how clusters activate and depend on each other, what operational constraints to plan for.	Sections 3, 2.3, 13 — Mandatory Base, Dependency Map, Constraints Framework
Technical Evaluator (CTO, Lead Engineer, Senior Architect)	Whether the architecture is technically sound, what the hard problems are, and where the honest gaps remain.	Sections 4–10, Appendix A — Cluster detail, Operational Case

GENERAL READER

If you are new to investigation AI architecture, start with the Start Here page on the next page, then read Section 1 (what Institutional Reasoning Engines are), then Section 11 (deployment profiles), then Appendix B (typology matrix). That is sufficient to evaluate whether this architecture is relevant to your organisation.

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Start Here — For Non-Technical Readers

THE PROBLEM THIS PAPER SOLVES

Every regulated investigation — financial crime, corporate fraud, HR misconduct, procurement — faces the same challenge: how do you prove that your AI-assisted findings are traceable to evidence, approved by a human, and untampered with? Most AI tools are either too generic to encode investigation methodology or too narrow to work across investigation types. This architecture solves that.

WHAT AN INSTITUTIONAL REASONING ENGINE IS

An AI system that forms a hypothesis, iterates through evidence, requires a human investigator to approve every finding before it becomes a record, and produces a tamper-evident audit trail from raw document to final finding. It is not a chatbot. It is not an automation tool. It is a professional investigation workflow encoded in software — with AI accelerating the analytical steps and humans accountable for every conclusion.

This Architecture IS	This Architecture IS NOT
A methodology encoding framework — investigation process as engineering constraints, not advisory policies.	A replacement for investigator judgment — every material output requires human approval.
A modular system that scales from 8 base components to a 35-component enterprise forensic platform.	A product available for purchase — this is an architectural blueprint.
Jurisdiction-agnostic — applicable to any regulated investigation in any industry.	A compliance guarantee in any specific jurisdiction — engage legal counsel for your context.
An AI acceleration layer with mandatory human gates at every material finding.	An autonomous AI investigation system — the human gate is architectural, not optional.
Tamper-evident by design at the base level; court-grade with Cluster G active.	Unconditionally immutable in all configurations — court-grade requires Cluster G.

THE ONE THING TO TAKE AWAY

Investigation methodology is encoded as engineering constraints — not as policies that humans might follow inconsistently. The technology serves the methodology. That inversion is what makes the output defensible.

How a Case Runs Through the System

An investigator opens a case and uploads the evidence — contracts, emails, bank records, interview recordings. The system makes every document searchable by meaning, maps the relationships between all entities, and runs an iterative reasoning loop that forms a hypothesis, retrieves supporting or contradicting evidence, and revises that hypothesis until confidence is earned or a gap is flagged. Every AI finding is a draft until the investigator approves it. Once approved, the finding is locked into a tamper-evident record. The final report cites every claim to its exact source document and page number. From case open to draft report: hours, not days.

Where to Go Next

- Decision Maker: Section 11 (Profiles) → Section 14 (Value) → Section 15 (Constraints)
- Implementation Lead: Section 3 (Mandatory Base) → Section 2.3 (Dependency Map) → Appendix A

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- Technical Evaluator: Sections 4–10 (Clusters) → Section 6.2 (Tiered LLM) → Section 7 (Reasoning Loop)
- First-Time Reader: Section 1 → Section 11 → Appendix B. Sufficient to evaluate relevance.

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1. Institutional Reasoning Engines — A New Category

The field of applied AI has produced two dominant archetypes: AI assistants that respond to queries, and AI automation tools that execute predefined tasks. Neither archetype is suited to investigation.

Investigation requires a third archetype — a system that:

- Maintains a working hypothesis that evolves as evidence accumulates.
- Iterates toward a defensible conclusion rather than answering a single query.
- Gates every material output through human judgment before it enters the evidentiary record.
- Produces a tamper-evident chain of reasoning from raw document to final finding — cryptographically secured by default, independently verifiable with Cluster G active.
- Operates within a defined methodology, not in service of a prompt.

This is an Institutional Reasoning Engine. It is institutionally grounded (the methodology is encoded, not improvised), reasoning-first (not retrieval-first or generation-first), and engine-class (it runs a defined process, not a conversation).

> **THE CATEGORY**
> An Institutional Reasoning Engine is not a chatbot, a search tool, or a document analyser. It is a professional reasoning workflow encoded in software — with AI accelerating the analytical steps and humans accountable for the conclusions.

1.1 Why Investigation Is the Beachhead

Investigation is the right first application because it has the clearest definition of what a correct output looks like (an STR, an HLIR, a case closure recommendation), the most demanding evidentiary standards (regulatory accountability, potential legal proceedings), and the most structured methodology (investigation frameworks, red flag typologies, reporting obligations). If the architecture works here, it works anywhere professional reasoning under uncertainty must produce a defensible output.

The pattern generalises: regulatory adjudication, compliance certification, credit risk determination, healthcare diagnosis audit. Investigation demonstrates the category. It does not define it.

1.2 The Six Unsolved Problems

Problem	Impact Without This Architecture
Document volume	100+ heterogeneous files per case — manual review takes days, not hours.
Reasoning depth	Multi-hop inference across transaction trails requires expert judgment AI cannot replicate linearly.
Network blindness	Entity relationships and circular flows are invisible to document-centric systems.
Hallucination risk	AI-generated findings without source citations are regulatorily indefensible.
Case isolation	Multi-investigator teams risk cross-case data leakage in shared AI systems.
Audit defensibility	No tamper-evident record of AI actions means investigation outputs cannot be trusted in proceedings.

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1.3 From Methodology to Architecture

Every architectural decision in this system traces directly to a specific investigation methodology requirement. This section makes those traces explicit — demonstrating that the technology serves the methodology, not the reverse.

Investigation Requirement	The Constraint It Imposed	The Engineering Decision
Every finding must be attributable to specific evidence — not to AI judgment.	The system cannot assert facts not recoverable from the case corpus.	Retrieval Verifier as a deterministic database lookup, not a probabilistic model check. Claims not recoverable from the corpus are stripped before reaching the investigator.
Human accountability is required at every material decision.	No AI output can enter the evidentiary record without a named human approving it.	Human Review Gate as a mandatory base component — not an optional workflow step. Approval is the record, not a step before the record.
Cases must not contaminate each other.	Shared AI memory across cases creates cross-contamination risk that access controls cannot reliably prevent.	Namespace isolation by architecture — one vector store per case. Cross-case contamination is structurally impossible, not policy-dependent.
The reasoning process must be auditable — not just the output.	Regulators and courts need to know what the AI retrieved, what it was told, what model version it used, and what a human approved.	Hash-chain immutability across every event — not a log file, a cryptographic chain. Tamper is mathematically detectable.
Findings must be reproducible.	Probabilistic model outputs create reproducibility risk across runs, model versions, and infrastructure.	Temperature zero for all extraction and finding tasks enables deterministic sampling within a run. Cross-run reproducibility is governed by model version pinning (B7), which stores exact model, version, and inference timestamp for every AI call.

These five traces are not the only methodology-architecture mappings in the system — they are the most fundamental. Every component in every cluster exists because the investigation methodology required something the prior generation of tools could not provide.

> **JURISDICTION NOTE**
> Legal and regulatory requirements referenced throughout this paper reflect general principles applicable across regulated investigation contexts. Specific obligations — admissibility standards, mandatory filing timelines, data protection requirements, AI governance rules — vary by jurisdiction. Engage qualified legal counsel to assess applicability in your specific regulatory environment before deploying any component of this architecture in production.

2. The Modular Architecture Framework

The architecture is structured in two tiers. The Mandatory Base is a set of eight components that every regulated investigation system must implement — without exception and without substitution. Above the base, seven modular clusters activate based on the investigation context: what evidence types are present, how complex the cases are, what institutional capabilities exist, and what the output will be used for.

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DESIGN PRINCIPLE

No cluster replaces a base component. Clusters add capability. The base establishes defensibility. An investigation system with sophisticated graph intelligence but no audit immutability is not a more advanced system — it is a less defensible one.

2.1 The Mandatory Base — 8 Universal Components

#	Component	What It Enforces
B1	Case-Scoped Isolation	Each case is a sealed evidence container. Cross-case contamination is prevented by architecture, not policy.
B2	Retrieval Verifier	Every AI claim is checked against the case corpus before reaching an investigator. Deterministic — not an AI judgment.
B3	Evidence Grounder	Source citation attached to every verified claim. Every finding is traceable to its evidence.
B4	Human Review Gate	No AI output enters the evidentiary record without explicit investigator approval. Human accountability at every finding.
B5	Audit Event Logger	Every system action recorded: queries, retrievals, AI calls, human decisions, model versions.
B6	Hash-Chain Immutability	Every log entry cryptographically linked. Any tampering anywhere in the chain is mathematically detectable.
B7	Model Version Pinning	Every case record stores exact model versions used. No silent upgrades mid-investigation.
B8	Algorithmic Bias Monitoring	Systematic over-flagging of demographic profiles is a regulatory liability. Active from the first case — not contingent on fine-tuning.

2.2 The Seven Modular Clusters

Cluster	Name	Activates When	Components
A	Documentary Evidence	Case corpus contains any documents — PDFs, spreadsheets, emails, contracts, reports.	7
B	Behavioural & Interview	Case involves recorded interviews, transcripts, witness statements, communication intercepts.	3
C	Entity & Network Intelligence	Investigation involves multiple entities whose relationships are material to the finding.	5
D	Advanced Reasoning	Case complexity requires iterative hypothesis formation across large or contradictory evidence sets.	5
E	Privacy Gateway	Any external AI API is used for reasoning — conditional mandatory the moment data leaves institutional boundaries.	3
F	Institutional Memory	Institution has sufficient historical case volume and governance maturity for fine-tuning.	2

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Cluster	Name	Activates When	Components
G	Integrity Elevation	Investigation outputs may be used in legal proceedings or regulatory enforcement actions.	2

2.3 Component Dependency Map

```
MANDATORY BASE (always active)
  B1 Case Isolation  |  B2 Retrieval Verifier  |  B3 Evidence Grounder
  B4 Human Review    |  B5 Audit Logger        |  B6 Hash Chain  |  B7 Model Pinning  |
B8 Bias Monitor

CLUSTER A — Documentary  (most common first activation)
  A1 Ingestion → A2 Chunking → A3 Embedding → A4 Extraction → A5 Templates → A6 Parallel
Gen → A7 Citation Map
  Feeds into: Cluster C (entities), Cluster D (reasoning), Cluster F (training)

CLUSTER B — Behavioural  (independent of A — can operate alone with Base)
  B1 Transcript Chunking → B2 Contradiction Detector → B3 Transcript Approval Gate
  Enhanced by: Cluster C (entity matching across transcript + documents)

CLUSTER C — Network  (requires A or B for entity source material)
  C1 Entity Extract → C2 Entity Resolution (HITL) → C3 Graph Build → C4 Graph Query → C5
Cross-Case Index
  Requires: A or B.  Feeds into: Cluster D (graph findings into reasoning loop)

CLUSTER D — Advanced Reasoning  (requires A; enhanced by B and C)
  D1 Tiered LLM → D2 Recursive Loop → D3 Hypothesis Tracker → D4 Evidence Scorer → D5
Sufficiency Check
  Requires: A.  Optional enhancement: B, C

CLUSTER E — Privacy Gateway  (conditional mandatory with any external API)
  E1 Pseudonymisation → E2 Token Map → E3 De-tokenisation Gate
  Activates: whenever external SOTA API is used in Cluster D

CLUSTER F — Institutional Memory  (requires A + D, minimum 200 cases)
  F1 Historical Fine-Tuning → F2 Negative Example Training
  Note: Algorithmic Bias Monitoring (formerly F3) promoted to Mandatory Base as B8

CLUSTER G — Integrity Elevation  (enhances Base B5/B6)
  G1 External Notarisation → G2 Transcript Chain of Custody
```

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3. Mandatory Base — 8 Universal Components

The eight base components are non-negotiable. They apply to every regulated investigation regardless of industry, jurisdiction, document type, AI capability level, or deployment scale. An investigation system that omits any one of these components cannot claim to produce a defensible, auditable output.

NON-NEGOTIABLE

These are not best practices or recommendations. They are minimum requirements. Any AI system used in a regulated investigation context that does not implement all eight base components is producing outputs whose defensibility cannot be assured.

#	Component	What It Enforces
B1	Case-Scoped Isolation	Each case is a sealed evidence container. Cross-case contamination is prevented by architecture, not policy.
B2	Retrieval Verifier	Every AI claim is checked against the case corpus before reaching an investigator. Deterministic — not an AI judgment.
B3	Evidence Grounder	Source citation attached to every verified claim. Every finding is traceable to its evidence.
B4	Human Review Gate	No AI output enters the evidentiary record without explicit investigator approval. Human accountability at every finding.
B5	Audit Event Logger	Every system action recorded: queries, retrievals, AI calls, human decisions, model versions.
B6	Hash-Chain Immutability	Every log entry cryptographically linked. Any tampering anywhere in the chain is mathematically detectable.
B7	Model Version Pinning	Every case record stores exact model versions used. No silent upgrades mid-investigation.
B8	Algorithmic Bias Monitoring	Systematic over-flagging of demographic profiles is a regulatory liability. Monitoring is mandatory from the first case processed.

3.8 B8 — Algorithmic Bias Monitoring

Algorithmic bias monitoring is a mandatory base component — not a Cluster F capability. This distinction matters: bias monitoring does not require historical fine-tuning to be active. It monitors whatever model is running, from day one. Any institution deploying AI in regulated investigation workflows carries regulatory and legal risk from systematic bias, regardless of case volume or model maturity.

What It Monitors	How Often	What Triggers Action
Flagged entity demographics vs base rate population — detecting if the system systematically over-flags entities of a particular type, geography, or profile.	Quarterly minimum — monthly for high-volume deployments.	Flag rate for any demographic segment deviating more than 15% from base rate triggers mandatory review before next investigation cycle.
Finding rates by case type, investigator, and model version — detecting drift across fine-tuning cycles.	After every fine-tuning deployment.	Any degradation in finding accuracy on holdout set, or increase in demographic skew, blocks deployment.
Human override patterns —	Monthly rolling review.	Systematic override pattern triggers

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What It Monitors	How Often	What Triggers Action
detecting if investigators systematically override AI findings in one direction for a particular entity type.		methodology review — either the model is biased or the investigators are.

MANDATORY FROM DAY ONE

Bias monitoring is active from the first case. An institution that waits until it has fine-tuned a model to begin monitoring has operated a biased system without oversight for its entire early deployment. The base component requirement means monitoring infrastructure is built before any case is opened.

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8. Hallucination Control and Evidence Grounding

An AI investigation system that generates findings without traceable source citations is not an investigation tool — it is a liability. The regulatory question is not whether AI produced the finding. It is whether you can show exactly which evidence supports every specific claim in the report.

REGULATORY REQUIREMENT

Without evidence grounding, every AI-generated finding is challengeable as unverifiable assertion. A single successful challenge can invalidate the entire investigation output. Evidence grounding is a legal requirement for investigation AI, not a product feature.

8.1 Three-Layer Grounding Architecture

```
Layer 1: RETRIEVAL CONSTRAINT
  Analyzer system prompt: 'You may only assert facts present in retrieved chunks.'
  'Any claim not supported by retrieved evidence must be tagged [UNVERIFIED]'
  Temperature: 0 for all extraction and finding tasks

Layer 2: RETRIEVAL VERIFIER  (deterministic — not AI-based)
  For every claim: does chunk_id exist in namespace? Is claim recoverable from chunk?
  Pass: attach citation.  Fail: strip claim, log to unverified register.
  This is a database lookup, not a model judgment.

Layer 3: EVIDENCE GROUNDER
  Attach to every verified claim:
    chunk_id | source_doc | page_ref | graph_node_ref | confidence_component
  Output: grounded_finding + citation_map + unverified_log
```

8.2 What Gets Stripped vs Flagged

Claim Type	Action	Consequence
Claim with valid chunk_id, retrieval-verified	Passes all layers	Included in grounded finding with citation.
Claim tagged [UNVERIFIED] by Analyzer	Stripped by Verifier	Moved to unverified register for human review.
Claim with chunk_id not found in namespace	Stripped + Alert	Potential hallucination — logged, escalated to investigator.
Finding with fewer than 3 distinct citations	Flagged — not stripped	Evidence sufficiency warning — section cannot auto-lock.
Graph claim without node/edge reference	Flagged	Requires graph query re-run to attach reference.
Transcript claim without speaker + timestamp	Flagged	Behavioural evidence requires full provenance chain.

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8.3 Citation Map in Final Report

The final report carries a Citation Map appendix — a complete index of every claim in the report matched to its source document, chunk ID, page reference, and graph reference where applicable. This appendix is automatically generated, locked into the final report, and included in the audit log. Any finding can be traced to its exact evidence source in under 30 seconds.

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SECTION 4 | CLUSTER A

Documentary Evidence

Activates when: the case corpus contains any documents

7 components

IN PLAIN LANGUAGE

Cluster A turns document piles into queryable evidence. Upload any file — a contract, a spreadsheet, an email chain, a scanned report — and the system makes every sentence retrievable by meaning, not just keyword. An investigator asking 'what did the subject know about this policy and when?' gets an answer in seconds, with the source document and page number attached. What it does not do: it does not reason about the documents — that is Cluster D. It does not map relationships between people and organisations — that is Cluster C. It makes documents searchable and citable. That is its job.

IN PRACTICE — HR MISCONDUCT INVESTIGATION

847 emails, 3 policy documents, and 14 months of calendar data uploaded. Investigator asks: 'What did the subject know about the code of conduct policy, and when?' System retrieves 4 email chunks and 1 policy acknowledgement record showing the subject received, opened, and responded to the updated policy 8 months before the incident. Total retrieval time: 6 seconds. Each result cited to document name, date, and page. Without Cluster A this query takes an analyst half a day.

4. Case-Scoped RAG Architecture

Retrieval-Augmented Generation grounds the AI's reasoning in actual case evidence. The critical design decision is namespace isolation: one vector collection per case, preventing cross-case contamination by architecture rather than policy.

4.1 Ingestion Pipeline

```
File Upload (Investigator)
       |
[Stage 1: Parser]    — PDF, Excel, CSV, email, images, audio transcripts -> clean text
       |
[Stage 2: Chunker]   — Document-type-aware segmentation
                          chunk_id | source_doc | page_ref | entity_tags | date_range
       |
[Stage 3: Embedder]  — Local embedding model (BGE-M3 or nomic-embed-text)
       |
[Stage 4: Indexer]   — Write to case-scoped namespace in Qdrant
                          Case_001_ns | Case_002_ns | Case_003_ns ...
       |
[Stage 5: Entity Extract] — Feed to Entity Resolution Layer -> then Graph Builder
```

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4.2 Document-Type-Aware Chunking

Document Type	Chunking Strategy	Critical Metadata
Bank statements / CSV	Row-window (50–100 rows), date continuity preserved	Date range, amount range, counterparty IDs
Emails / communications	Per-email chunks	Sender, recipient, timestamp, thread ID
KYC / identity documents	Field-level extraction	Entity name, ID type, verification status
Contracts / agreements	Clause-level chunks	Section header, party names, effective date
Investigation notes	Paragraph-level	Author, timestamp, linked entities
Call recordings / transcripts	Speaker-turn level	Speaker ID, timestamp, session ID, interviewer ID, emotion markers if available
Video statement transcripts	Speaker-turn + segment	Speaker ID, timestamp, cross-reference flags to prior statements

4.3 Retrieval Performance Benchmarks

Operation	Target Latency	Notes
100-page PDF ingestion	45–90 seconds	Async — investigator not blocked
50K row transaction CSV	2–4 minutes	Async background job
Query → retrieval (post-index)	< 1 second	Namespace-filtered semantic search
Graph query (entity network)	< 2 seconds	Local graph DB instance
Retrieval verification check	< 0.5 seconds per claim	Deterministic lookup against case namespace
Full report (parallel sections)	60–90 seconds	All sections simultaneously after corpus indexed

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SECTION 5 | CLUSTER B

Behavioural & Interview

Activates when: the case involves recorded interviews, transcripts, or witness statements

3 components

IN PLAIN LANGUAGE

Cluster B does one thing that no document system can: it detects when someone says something in an interview that contradicts what the evidence shows. A subject claims they never worked with a particular supplier — the system checks every document and transaction record in the case and flags the contradiction automatically, citing both the statement and the contradicting evidence. What it does not do: it does not judge whether the contradiction is significant — the investigator does. It surfaces the contradiction with citations on both sides. The investigator decides what it means.

IN PRACTICE — PROCUREMENT FRAUD INTERVIEW

Subject states in interview session 2: 'I had no prior relationship with Pinnacle Supplies before they were onboarded.' Contradiction Detector cross-references against case corpus. Flags: subject's personal email appears in Pinnacle's incorporation documents filed 14 months earlier (Cluster A chunk reference: CORP_REG_003, page 2). Both the statement and the contradicting document are cited side by side. Finding generated for investigator review with separate transcript approval gate. Investigator did not need to search for the contradiction — the system surfaced it.

9. Training on Historical Data and Transcript Intelligence

9.1 Fine-Tuning on Institutional Case Data

The system can be trained on the institution's own historical investigation data — both raw transaction records and completed investigation reports (STRs, HLIRs, case closure documents). This gives the Agent LLM an institutional memory: it learns what constitutes a finding in this specific context, what red flag patterns this institution has historically recognised, and what the reporting standard looks like for this regulator and jurisdiction.

INSTITUTIONAL MEMORY

A system fine-tuned on your own case history will outperform a generic model on your specific fraud typologies. The model learns your patterns, your terminology, your reporting standards. This is the compounding advantage of building rather than buying.

Fine-Tuning Data Requirements

- Source material: pseudonymised historical STRs, HLIRs, investigation notes, transaction datasets linked to confirmed fraud cases, and case closure reports.
- Pseudonymisation is mandatory before any data leaves the case management environment for training. Token maps from training data are destroyed after the fine-tuning run.
- Negative examples are as important as positive ones: closed cases with no finding teach the model what does not constitute suspicious activity in this context.
- Volume: a minimum of 200–500 completed cases produces meaningful fine-tuning signal. Below this threshold, general model capability is preferable to fine-tuning on insufficient data.

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Fine-Tuning Governance

- The fine-tuned model receives a new model_id and model_version. It is treated as a new model for all case record purposes.
- Cases opened before the fine-tuning run are not retroactively affected — they retain their original model version in the audit record.
- Fine-tuning is periodic (quarterly or semi-annually), not continuous. Continuous fine-tuning creates model drift that is difficult to audit.
- A holdout evaluation set from historical cases is used to validate the fine-tuned model before production deployment.

9.2 Transcript Intelligence for Behavioural Investigations

Behavioural investigation cases — involving interview recordings, call intercepts, or statement transcripts — require capabilities beyond document analysis. A subject's spoken words, contradictions between statements and documentary evidence, and communication patterns across multiple recorded interactions are investigative signals that the standard RAG pipeline was not designed to surface.

> **TRANSCRIPT CAPABILITY**
> Transcripts resolve ambiguities that documents cannot. When a subject states in a recorded interview 'I have no relationship with that merchant' and the transaction graph shows 47 edges between them, the contradiction is itself a material finding. The architecture can detect and cite it.

Transcript Processing Pipeline

```
Raw Recording / Transcript File
       |
[Transcription]           — If audio: speech-to-text (local model, on-premise)
                            Output: time-stamped, speaker-diarised transcript
       |
[Chunker: Speaker-Turn]  — Chunk by speaker turn, not paragraph
                            Metadata: speaker_id | timestamp | session_id |
interviewer_id | emotion_markers
       |
[PII Scrubbing]           — Extended Presidio recognisers for spoken-form names,
                            informal account references, colloquial location markers
       |
[Index to Case Namespace] — Transcript chunks enter same Qdrant namespace as documents
       |
[Contradiction Detector]  — Cross-reference: transcript claims vs documentary evidence
                            vs graph edges. Flag discrepancies to Planner.
```

Contradiction Detection — Types and Outputs

Contradiction Type	Example	Output
Statement vs transaction record	Subject denies relationship with merchant. 47 transaction edges exist.	Contradiction finding: statement [chunk_id] vs graph edges [node_ref]. Material.

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Contradiction Type	Example	Output
Statement vs document	Subject claims first contact in 2024. Email from 2022 in corpus.	Contradiction: transcript [chunk_id] vs email [chunk_id]. Date discrepancy cited.
Statement vs prior statement	Subject gives different account ownership story across two interviews.	Inconsistency: Session 1 [chunk_id] vs Session 3 [chunk_id]. Recommend clarification.
Statement vs KYC	Subject describes business as retail. KYC shows industrial classification.	Discrepancy: transcript [chunk_id] vs KYC field [chunk_id]. Flag for investigation.

Sensitive Evidence Handling for Transcripts

- Transcript-derived findings carry an additional human approval flag before entering the final report — separate from standard section approval.
- Recording transcripts contain unstructured PII at high density. Extended PII recognisers are required for spoken-form identifiers before any transcript chunk enters the pseudonymisation pipeline.
- Emotion and hesitation markers (if available from the recording system) are preserved as metadata, not stripped. They are investigative signal in behavioural cases.
- Chain of custody for recordings: original file hash is logged at ingestion. Transcript is linked to original by hash reference in the audit chain.

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SECTION 6 | CLUSTER C

Entity & Network Intelligence

Activates when: investigations involve multiple entities whose relationships are material to the finding

5 components

IN PLAIN LANGUAGE

Cluster C answers the question no document search can: who is connected to whom, and does that network reveal something the individual documents do not? Six independently filed insurance claims look unrelated until a graph shows that four of them share two witnesses. A procurement investigation looks like one corrupt manager until a graph shows three suppliers were all registered by the same agent within six months of each other. What it does not do: it does not interpret what the network means — it surfaces the pattern and cites the evidence. The investigator determines whether the connection is coincidental or material.

IN PRACTICE — INSURANCE FRAUD RING

Six insurance claims opened independently across three adjusters over 11 months. No single adjuster saw more than two claims. Graph query on shared entity attributes surfaces: 4 of the 6 claims share 2 witnesses; 3 claims name the same repair contractor; 2 claims reference the same medical examiner. Cross-case index flags all 6 claims to a senior investigator. None of this was visible from any individual claim file. The network was only visible from above.

3. Entity Resolution Layer

Entity resolution is the hardest practical problem in the architecture — and the one most commonly skipped in design documents. Before the graph layer can function, you must determine that 'Rajesh Kumar' in the bank statement, 'R. Kumar' in the KYC document, and 'RJSH_KMR' in the transaction log are the same node. Skipping this step does not make the problem disappear — it produces a fragmented graph where the circular flow you need to detect never surfaces because the nodes are duplicated.

CRITICAL DEPENDENCY

Entity resolution quality directly determines graph quality. A 10% entity duplication rate in a 500-node graph produces 50 phantom nodes — enough to break circular flow detection entirely. This is the gap that most investigation AI systems fail to address honestly.

3.1 Three-Tier Resolution Pipeline

```
Document Parser Output (raw entities)
       |
[TIER 1: Exact Match]
  Fields: National ID · Account number · Registered mobile · Tax ID
  Action: Auto-resolve -> merge nodes -> log merge decision to audit chain
  HITL:   Notification only — investigator can override within 24hrs
       |
[TIER 2: Fuzzy Match]
```

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```
  Fields: Name variants · Address overlap · Transliterations · Phonetic similarity
  Action: System proposes merge with confidence score + supporting evidence
  HITL:   BLOCKING — investigator must Approve / Reject / Defer before graph builds
       |
[TIER 3: Contextual Match]
  Fields: Shared director · Same registered address · Overlapping counterparties
          Corroborating transcript evidence · Common communication patterns
  Action: System flags as 'possible alias' with supporting evidence list
  HITL:   BLOCKING — investigator decides: merge / separate / link as alias edge
```

3.2 Human-in-the-Loop Confirmation Design

Tier	Confidence	HITL Mode	Audit Record
Tier 1 — Exact	95–99%	Non-blocking notification. Override window: 24hrs. Default: proceed.	Merge decision + field matches + override flag if changed
Tier 2 — Fuzzy	60–94%	Blocking confirmation screen. Present: evidence for merge, evidence against, confidence score. Investigator chooses.	Decision + evidence presented + investigator ID + timestamp
Tier 3 — Contextual	<60%	Blocking flag. Present: contextual overlaps, possible alias hypothesis. Three options: merge / separate node / alias edge.	Full evidence set + decision rationale + investigator ID
No match	N/A	Entities remain separate. System flags if they share future transaction edges.	Flag timestamp + triggering evidence

3.3 Resolution Decisions as Investigation Findings

Entity resolution decisions are not administrative housekeeping — they are investigation findings. The determination that two entities are the same person may itself be a material finding in a case. Every resolution decision is therefore:

- Logged immutably in the audit chain with full evidence provenance.
- Reversible by a senior investigator with a documented justification.
- Visible in the case hypothesis tracker as a confirmed or possible entity link.
- Cited in the final report if the merged identity is material to the finding.

3.4 Transcript Evidence in Entity Resolution

Behavioural investigation cases provide resolution signals that documentary evidence cannot. A subject who mentions 'my Axis account ending 4782' in a recorded interview resolves an ambiguous account reference in the transaction data. The resolution pipeline accepts transcript chunks as supporting evidence in Tier 2 and Tier 3 decisions, with the transcript source cited in the audit record.

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5. Graph Intelligence Layer

Document-centric systems are blind to the network structures that define financial crime. Money laundering, counterparty collusion, and transaction layering are fundamentally graph phenomena. They cannot be reliably detected by reading documents — they must be traversed as connected structures.

> **WHY GRAPH IS NOT OPTIONAL**
> A system without graph analysis can identify that unusual transactions occurred. Only a system with graph analysis can identify that these three merchants form a circular network processing funds from a common originating wallet. The difference is the difference between an alert and a finding.

5.1 Four Financial Crime Patterns Requiring Graph Analysis

Pattern	Document Detection	Graph Detection
Circular flows	Missed — transactions appear legitimate in isolation	Detected: closed loop across 3+ nodes identified by path traversal
Counterparty clustering	Missed — no cross-entity view	Detected: multiple subjects sharing a small set of counterparties
Layering chains	Partial — only if all documents are in same case	Detected: sequential hops across N entities, weighted by amount and velocity
Mule network detection	Missed	Detected: high-degree nodes with abnormal edge weight distribution

5.2 Graph Architecture

```
Entity Resolution Output (resolved nodes)
       |
[Graph Builder]    — Entities -> Nodes | Transactions -> Edges (weighted: amount,
frequency)
       |
[Neo4j / Memgraph] — Case-scoped graph (same isolation as Qdrant namespace)
       |
+-----------------------------------------------+
|           GRAPH QUERY LAYER                   |
| Circular flow detection (Cypher path queries) |
| Community detection (graph algorithms)        |
| Shortest path between flagged entities        |
| Anomalous edge weight / velocity detection    |
| Contradiction check: transcript vs transaction|
+-----------------------------------------------+
       |
[Graph Findings]   — Structured output fed to Planner alongside RAG retrieval
```

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5.3 Cross-Case Graph Governance

The cross-case graph layer surfaces fraud rings across independently opened cases — the single most powerful capability in the architecture. It also introduces the most significant governance complexity and must be explicitly controlled.

Governance Dimension	Design Decision
Legal basis	Cross-case analysis operates only within the institution's investigation mandate. It is not general surveillance. Scope is defined by the investigation authority document before deployment.
What crosses case boundaries	De-identified entity tokens only. Real identifiers never cross case boundaries. The cross-case index contains tokens and graph topology — no names, IDs, or amounts.
What a cross-case query produces	A lead, not a finding. The output is: 'Token ENTITY_T047 appears in 3 other cases.' This triggers a new investigation or case linkage decision — it does not auto-merge cases.
Access control	Senior investigators and compliance officers only. Not available to standard case workers. Access is logged per query.
Audit record	Both the querying case and the matched cases receive an audit log entry recording: query timestamp, approver ID, token matched, action taken.

5.4 Recommended Stack

- Neo4j Community Edition: mature Cypher query language, strong Python SDK, self-hostable, best for most teams.
- Memgraph: higher performance for real-time streaming graph updates. Better fit if transaction events feed continuously rather than in batch.
- Both are open-source and self-hosted, maintaining data residency compliance in any jurisdiction.

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SECTION 7 | CLUSTER D

Advanced Reasoning

Activates when: case complexity requires iterative hypothesis formation

5 components

IN PLAIN LANGUAGE

Cluster D is the reasoning engine. The other clusters find and organise evidence — Cluster D thinks about it. It forms a hypothesis, searches for evidence that tests the hypothesis, challenges its own reasoning, revises the hypothesis, and repeats until it has either earned a confident finding or identified a gap the investigator needs to fill. A 340-page regulatory document set that would take an analyst three days to synthesise produces a draft finding in 90 seconds. What it does not do: it does not reach conclusions — it drafts findings for investigator review. Every output is a draft until a human approves it. The investigator is not reviewing AI output — they are reviewing a draft that they are accountable for.

IN PRACTICE — REGULATORY EXAMINATION PREPARATION

340-page policy framework, 12 audit reports spanning 3 years, and 6 board meeting minutes uploaded. Investigator asks: 'Does our actual practice match our stated policy on outsourcing risk?' Iteration 1: hypothesis formed, policy requirements retrieved. Iteration 3: board minute from 18 months ago contains resolution that contradicts current policy document — hypothesis revised. Iteration 5: finding locked at evidence score 0.91, citing 9 distinct source documents. Draft report section generated in 74 seconds. An analyst reading sequentially would have taken 3 days to find the same contradiction.

6. Tiered LLM Orchestration

Not all investigation tasks require the same reasoning capability. The tiered architecture assigns compute to capability level, reserving frontier models exclusively for decisions that precede human approval gates. This simultaneously controls cost, data exposure risk, and latency.

6.1 The Three-Tier Model

Tier	Model Class	Deployment	Role
Orchestration	8B–14B (local)	On-premise	Task routing, queue management, status updates, form population, structured extraction with schema validation
Agent	70B (local)	On-premise	Document analysis, entity extraction, evidence summarisation, hypothesis generation, graph query translation, contradiction detection
Reasoning	SOTA API (pseudonymised)	External	Final risk narrative, anomaly judgment, HLIR/SLIR synthesis — fires at human gates only

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6.2 Structured Extraction Layer

LLM reasoning is probabilistic. The same input can produce different extractions across runs. For investigation pipelines, this is a reliability problem. The structured extraction layer sits between LLM output and the graph builder, enforcing a validation schema before any entity becomes a graph node.

- Every entity extraction passes through a schema validator: required fields, expected formats, confidence floor.
- Malformed, incomplete, or sub-threshold extractions are flagged for human review rather than silently entering the graph.
- Temperature is set to zero for all extraction tasks — deterministic sampling, not creative generation.
- Extraction results are versioned alongside the model version that produced them.

6.3 SOTA Activation Gates

The SOTA model activates at exactly three points in the investigation lifecycle — all preceding a human decision:

- Final risk narrative generation before case closure review.
- Anomaly reasoning synthesis when the 70B agent flags pattern uncertainty.
- HLIR/SLIR draft generation for investigator sign-off.

> **COST CONTROL**
> At 200 cases/month, SOTA API fires approximately 600 times. Conservative estimate including longer contexts, retries, and parallel section generation: $500–$1,500/month. Even at the upper bound, this is under 2% of total platform cost and less than the benefit of a single avoided regulatory penalty.

6.4 Model Version Pinning

Every case record stores the exact model versions used at each reasoning step: model_id, model_version, temperature, and inference_timestamp for every AI call. Model upgrades require a case migration protocol — open cases are never silently transferred to new model versions mid-investigation. This is a regulatory defensibility requirement, not an operational preference.

7. Recursive Reasoning Architecture

Standard RAG answers the question you ask. Recursive RAG discovers the questions you should have asked. The recursive reasoning agent replicates the cognitive process of an experienced investigator: form a hypothesis, retrieve evidence, analyse, verify every claim, challenge the hypothesis, revise, repeat — until confidence is earned or a gap is surfaced for human review.

7.1 The Reasoning Loop

```
Investigation Agent initialised with:
  Case State + Goal + Framework + Model Version + Prior Findings

Loop:
```

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```
  +-- PLAN     -> Planner queries: What evidence do I need next? (RAG + Graph)
  +-- RETRIEVE -> Execute RAG query + Graph query against case namespace
  +-- ANALYZE  -> What does this evidence tell me? (70B agent, temp=0)
  +-- VERIFY   -> Can every claim be retrieved from the corpus? [DETERMINISTIC]
  +-- GROUND   -> Attach chunk_id + source_doc + page_ref to every verified claim
  +-- CRITIQUE -> Logical consistency + gap identification
  +-- SCORE    -> Update calibrated evidence score. Threshold met? Stop. Else loop.

Termination: Evidence score threshold met | Iteration ceiling | Corpus exhausted
             | Evidence sufficiency check fails | Human interrupt
```

7.2 The Seven Agent Components

Investigation State Object

Persistent memory across iterations. Carries all findings, gaps, queries fired, calibrated scores, identified entities, evidence citations, graph findings, and the evolving working hypothesis. Every model call receives the complete state. Model version is stored in state and cannot change mid-session.

Planner

Receives current state and outputs exactly one next action: a RAG query, a graph query, a contradiction check, or a stop signal. Does not generate analysis or write findings. Separation of planning from analysis prevents circular reasoning.

Analyzer

Receives retrieved chunks and graph findings. Generates section-specific analysis in retrieval-only mode: it cannot assert facts not present in retrieved evidence. Every output claim carries a provisional chunk_id tag. Temperature zero.

Retrieval Verifier

Deterministic — not probabilistic. For every claim in the Analyzer output: checks that the cited chunk_id exists in the case namespace and that the claim is recoverable from that chunk. Claims that fail verification are stripped before the Critic sees them. Unverifiable claims are logged in the unverified assertions register for human review.

Evidence Grounder

Attaches the verified chunk_id, source_doc, page_ref, and graph node/edge reference to every claim that passed retrieval verification. Produces the grounded finding and the citation map for that iteration.

Critic

Evaluates grounded, retrieval-verified output across three dimensions only: logical consistency with prior findings, gap identification, and iteration recommendation (continue / stop / escalate to human). The Critic does not evaluate hallucination — the Retrieval Verifier handles that.

Hypothesis Tracker

Maintains and sharpens the working theory across iterations. Hypothesis updates are only accepted if backed by verified, grounded citations or confirmed graph findings. The tracker rejects hypothesis updates not supported by the evidence — preventing the system from reasoning beyond what the corpus supports.

Iteration	Hypothesis	Evidence Source	Score
1	Unusual transaction volume — possible fraud	RAG: transaction CSV chunks	0.41
2	Related party flows — possible	RAG: KYC + Graph: counterparty	0.63

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Iteration	Hypothesis	Evidence Source	Score
	layering	cluster	
3	Structured layering: 3 merchants, circular flow	Graph: path traversal + RAG: invoice analysis	0.81
4 LOCKED	Confirmed circular flow. PMLA S.3 exposure. Recommend STR.	Graph + RAG + Transcript contradiction	0.91

7.3 Termination Controller

- Calibrated evidence score meets section threshold (0.80 standard / 0.90 final findings).
- Iteration ceiling reached (3 simple / 5 medium / 7 complex).
- Corpus exhausted — no new evidence in last iteration.
- Evidence sufficiency check fails — citations below minimum density. System flags: 'High confidence on thin evidence — manual review required.'
- Human interrupt — investigator redirects or terminates.

EVIDENCE SUFFICIENCY

High confidence on thin evidence is a red flag, not a green light. Minimum citation density: 3+ distinct source chunks for any material finding. A score threshold alone is insufficient — both conditions must be met before a section locks.

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SECTION 8 | CLUSTER E

Privacy Gateway

Activates when: any external AI API is used for reasoning

3 components

IN PLAIN LANGUAGE

Cluster E is the data protection layer for institutions that use external AI services. It replaces every personal identifier in the case data with an anonymous token before anything leaves the institution's systems. The AI reasons about tokens, not people. The real identifiers are restored only in the final report, only after an investigator approves, and only inside the institution's own environment. What it does not do: it does not make external AI use unconditionally safe. Pseudonymisation controls what enters the AI — it does not fully control what the AI infers. Legal counsel must assess whether the reasoning output constitutes personal data under applicable law before treating this cluster as a complete privacy control.

IN PRACTICE — FINANCIAL CRIME, EXTERNAL AI REASONING

Case corpus contains KYC records and transaction data for a suspected layering scheme. Institution uses SOTA external API for final risk narrative generation. Before any data leaves the institution: account numbers replaced with ACCT_T089, ACCT_T112; individual names replaced with IND_T034, IND_T041; company names replaced with ENT_T007. External model reasons on tokens. Output: 'ACCT_T089 transferred to ENT_T007 on 6 occasions, consistent with layering pattern.' Investigator approves. De-tokeniser restores real identifiers in final report only. External API never processed a real name or account number.

10. Privacy-Preserving Pseudonymisation

Every external API call introduces data residency and regulatory exposure. The pseudonymisation layer makes re-identification impossible without the token map — which never leaves the on-premise environment. This is the control that enables use of frontier AI reasoning capabilities without transferring personal data outside institutional boundaries.

10.1 The Pseudonymisation Pipeline

```
Raw Case Data (full PII)
        |
[Entity Extractor]      — Local NER + regex (Presidio + custom recognisers)
        |
[Token Map Store]       — In-memory, session-scoped, encrypted at rest, never leaves
environment
        |
[Pseudonymised Payload] — Amounts, dates, transaction types, patterns intact
                          Names, IDs, account numbers replaced with consistent tokens
        |
[External SOTA API]     — Reasoning on pseudonymised data only
        |
```

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```
[Human Checkpoint]      — Investigator reviews tokenised reasoning + citation map
       | [APPROVE]
[De-tokeniser]          — Restores real identifiers in final report only
       |
[Audit Log]             — Records: pseudonymisation event + approval + de-tokenisation
timestamp
```

10.2 Field Treatment

Field Type	Treatment	Rationale
Names, national IDs, mobile numbers	Replace with consistent token	Direct PII — highest re-identification risk
Account / wallet / merchant IDs	Replace with consistent alias	Traceable to identity
Transaction amounts	Retain	Required for pattern reasoning
Timestamps and dates	Retain	Required for chronology analysis
Transaction types and channels	Retain	Required for behavioural analysis
City / region geography	Retain	Risk signal; not pin-point locator
Specific addresses	Generalise to region level	Pin-point location risk
Spoken-form references in transcripts	Extended recogniser — token	High-density informal PII

DATA RESIDENCY NOTE

Pseudonymised data transmitted to an external API retains its character as personal data if re-identification is possible under applicable law. Legal defensibility requires: (1) token map on-premise only, (2) no combination of retained fields enabling re-identification, (3) documented data processing agreement with API vendor. Engage legal counsel before first external API call in any production deployment.

OUTPUT RE-IDENTIFICATION RISK — KNOWN LIMITATION

Pseudonymisation controls what enters the external AI — it does not control what the AI reasons about or what its output implies. A reasoning model given pseudonymised data may produce outputs whose content, when combined with other information available within the institution, enables re-identification of a data subject. This is inference re-identification risk and is a known limitation of pseudonymisation as a privacy control. Practical mitigations: (1) The human review gate is the primary operational control. (2) AI reasoning outputs should be reviewed by legal counsel before the pseudonymisation gateway is treated as a complete privacy control. (3) Outputs that carry re-identification risk should not be stored verbatim — investigators should synthesise the finding in their own words. Cluster E is a strong control, not a complete one. Privacy governance for AI-assisted investigation requires ongoing legal review, not a one-time architectural decision.

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SECTION 9 | CLUSTER F

Institutional Memory

Activates when: the institution has sufficient historical case volume for fine-tuning

2 components

IN PLAIN LANGUAGE

Cluster F makes the system smarter over time by training it on the institution's own completed investigations. After 200 or more cases, the model learns what a finding looks like in this specific context — your fraud typologies, your reporting standards, your regulatory environment. It learns what you have historically flagged and, equally importantly, what you have historically cleared. What it does not do: it does not train continuously or automatically. Training runs are deliberate, governed, and evaluated against a holdout set before deployment. A model that performs worse than the general model on the holdout set is not deployed, regardless of how much training data was used.

IN PRACTICE — INSTITUTIONAL MEMORY AT 18 MONTHS

After 300 completed cases, first fine-tuning run. The fine-tuned model learns a trade-based fraud typology specific to this institution's merchant portfolio — over-invoicing on cross-border goods where the margin spread exceeds 40%. This pattern does not appear in any generic training data. On the next case involving the same merchant category, the model flags the spread in iteration 1 rather than iteration 4. Investigation time: 4 hours vs 11 hours on the same typology before fine-tuning. The institution's accumulated experience is now encoded in the model.

Cluster F — Institutional Memory

F1–F2: Historical Fine-Tuning and Negative Example Training

When an institution has accumulated sufficient case history — minimum 200 completed investigations — the agent model can be fine-tuned on that institutional data. This builds institutional memory: the model learns what constitutes a finding in this specific context, what typologies this institution has historically recognised, and what the reporting standard looks like.

- F1 — Historical Case Fine-Tuning: Train the agent model on pseudonymised completed cases. Periodic (quarterly/semi-annual), not continuous. New model version ID assigned after each run — silent upgrades prohibited.
- F2 — Negative Example Training: Include closed cases with no finding. Teaches the model what does not constitute suspicious activity in this environment. Without negative examples, the model learns only what fraud looks like — not what it does not look like.

NOTE ON B8

Algorithmic Bias Monitoring (previously F3) has been moved to the Mandatory Base as B8. It is required from the first case processed — not from the point at which fine-tuning begins.

F1.1 — Fine-Tuning Evaluation Protocol

Fine-tuning carries regulatory risk — a poorly governed training run can introduce systematic bias or degrade investigation quality. The following protocol is mandatory before any fine-tuned model is deployed to production.

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Step	Requirement	Pass Condition	Failure Action
Holdout set preparation	Reserve 20% of historical cases before training. Never used in training. Minimum 40 cases at 200-case threshold.	Holdout set representative across case types and outcomes (finding vs no-finding split within 10% of full dataset).	Extend training data collection before proceeding.
Bias pre-check	Run B8 bias audit on proposed training data before training begins.	No demographic segment over-represented by more than 15% relative to base rate.	Remediate training data composition. Do not proceed with skewed training set.
Performance evaluation	Run fine-tuned model against holdout set. Measure case-level finding accuracy.	Fine-tuned model matches or exceeds general model performance on holdout set.	Root cause analysis required. Do not deploy.
Bias post-check	Re-run B8 audit on fine-tuned model outputs against holdout set.	No new demographic skew introduced by fine-tuning.	Block deployment. Audit training data for systematic labelling bias.
Model version assignment	Assign new model_id to fine-tuned model.	New model_id logged in deployment record.	N/A — administrative step.
Cadence	Quarterly maximum.	No exceptions.	Override requires written sign-off from governance head and compliance.

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SECTION 10 | CLUSTER G

Integrity Elevation

Activates when: investigation outputs may be used in legal proceedings or regulatory enforcement

2 components

IN PLAIN LANGUAGE

Cluster G answers the hardest question an investigation record faces in a courtroom or regulatory proceeding: how do you know this was not altered after the fact? Without this cluster, the answer is 'trust us.' With it, the answer is 'verify it independently.' Daily external anchoring of the audit chain means a third party can mathematically confirm that every entry is exactly as it was when it was created. What it does not do: it does not replace legal admissibility review. Evidence integrity is a necessary condition for admissibility — it is not sufficient. Legal counsel must confirm admissibility requirements in the relevant jurisdiction.

IN PRACTICE — ENFORCEMENT ACTION, 14 MONTHS LATER

Investigation closes. 14 months later, enforcement proceedings begin. External counsel requests the complete audit chain. Hash verification runs in 4 seconds: 247 audit entries, chain intact, no tampering detected since case close. Daily notarisation anchors confirm chain state at 14 independent time points. Opposing counsel challenges the integrity of the AI-generated findings. Institution responds: the chain is independently verifiable. The integrity is not asserted — it is mathematically provable.

Cluster G — Integrity Elevation

The mandatory base (B5+B6) produces institutional-grade audit integrity — sufficient for internal investigations, regulatory examinations, and most compliance contexts. When investigation outputs may be used in legal proceedings, enforcement actions, or external arbitration, two additional components elevate that integrity to court-grade.

- G1 — External Notarisation: Daily anchor of the audit chain hash to an external timestamp authority or QLDB-style ledger. Provides cryptographic proof of existence at a point in time that is independent of the institution's own systems.
- G2 — Transcript Chain of Custody: Original recording file hash linked to transcript in the audit chain. Proves the transcript has not been altered from the source recording.

COURT-GRADE INTEGRITY

With G1 and G2 active, the audit chain is independently verifiable. The institution does not need to prove the chain's integrity — a third party can verify it. This is the difference between 'we say it was not altered' and 'it is mathematically provable that it was not altered.'

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11. Calibrated Evidence Scoring

Raw model confidence is not well-calibrated: a 70B model can output 0.94 confidence on a hallucinated finding. This section replaces raw confidence with a deterministic, auditable composite score that cannot be gamed by a miscalibrated model.

11.1 The Four-Component Evidence Score

Component	What It Measures	How Computed	Weight
Citation Density	Are material findings supported by multiple independent sources?	Number of distinct chunk_ids supporting the finding (floor: 3 for material findings)	30%
Retrieval Verification Rate	What proportion of claims survived the deterministic verifier?	Verified claims ÷ total claims in Analyzer output	30%
Framework Coverage	What proportion of the investigation checklist is addressed?	Checklist items with grounded evidence ÷ total applicable items	25%
Cross-Source Consistency	Do independent sources agree or contradict?	Agreement rate across document types (transaction vs KYC vs comms vs transcript)	15%

```
Evidence Score =
  (0.30 x Citation Density Score)
+ (0.30 x Retrieval Verification Rate)
+ (0.25 x Framework Coverage Score)
+ (0.15 x Cross-Source Consistency Score)

Threshold: 0.80 for section lock  |  0.90 for final finding lock
Override:  Evidence sufficiency check must also pass (min 3 citations per material
claim)
Note:      Raw model confidence is logged as metadata but does not gate the score.
```

11.2 What This Fixes

- A model that is highly confident about a finding supported by one document chunk will score low on Citation Density and fail the evidence sufficiency check regardless of confidence.
- A model that retrieves extensively but asserts beyond retrieved evidence will score low on Retrieval Verification Rate.
- A finding that addresses only half the investigation framework will score low on Framework Coverage regardless of how well-cited the addressed sections are.
- The score is reproducible: the same evidence corpus will produce the same score on re-run. This makes it auditable and defensible.

> **ON THE WEIGHTS**
>
> The 30/30/25/15 distribution is an illustrative starting point, not a specification. Citation density and retrieval verification rate are weighted equally and highest because they are deterministic — they

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> cannot be inflated by a miscalibrated model. Framework coverage and cross-source consistency involve model judgment and require calibration against the institution's specific investigation methodology. Treat these weights as a hypothesis to be tested against your own holdout set, not a fixed configuration to be implemented.

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12. Audit Immutability

An investigation system without a tamper-evident audit trail is not an investigation system. For AI-generated outputs to be defensible in regulatory proceedings, enforcement actions, or litigation, every AI action must be provably unaltered from the moment it was generated to the moment it is presented as evidence.

> **LEGAL REQUIREMENT**
> Regulators and courts will ask: how do you know the AI output was not modified after generation? Without an immutable audit chain, you cannot answer this question. Audit immutability is the difference between an admissible investigation record and a challenged one.

12.1 Complete Event Taxonomy

Event Category	Logged Elements
File ingestion	file_name, file_hash, ingestion_timestamp, chunk_count, namespace_id
Transcript ingestion	file_hash of original recording, transcription_model, speaker_count, session_id
Entity resolution	decision tier, entities_presented, evidence_shown, decision, investigator_id, timestamp
RAG retrieval	query_text, top_k_chunks, chunk_ids, retrieval_timestamp, namespace_id
Graph query	cypher_query, result_nodes, result_edges, query_timestamp
Structured extraction	model_id, schema_version, extraction_output_hash, validation_result
Retrieval verification	claims_count, verified_count, stripped_count, unverified_register_hash
Evidence score computation	component_scores, composite_score, threshold_met, iteration_number
Hypothesis update	prior_hypothesis, updated_hypothesis, triggering_evidence_ids, timestamp
Human review action	investigator_id, section_id, action, diff_hash, timestamp
De-tokenisation event	approver_id, case_id, timestamp, report_hash
Final report generation	report_hash, citation_map_hash, model_versions_used, timestamp

12.2 Hash-Chain Design

```
Each log entry:
{
  event_id:    UUID
  timestamp:   ISO 8601
  event_type:  [ingestion | resolution | retrieval | ai_call | approval | ...]
  payload:     { event-specific fields }
  payload_hash: SHA-256(payload)
  chain_hash:  SHA-256(payload_hash + prior_chain_hash)
}

Verification: recompute chain_hash for each entry.
```

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```
Any modification anywhere breaks all subsequent hashes.
Tamper is cryptographically detectable.

Optional enhancement: daily external notarisation of chain hash
  via timestamp authority or QLDB-style ledger anchor.
  Elevates from institutional-grade to court-grade evidence integrity.
```

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13. Draft Report Generation

The integrated pipeline produces a structured draft investigation report in 60–90 seconds, with every claim retrieval-verified, source-cited, and calibrated-scored. All outputs are gated by human approval before entering the final record.

13.1 Query Template Architecture

Section	RAG Query	Graph Query
Subject Background	Summarise entity profile, KYC details, registration information	Entity node properties, director connections, registration date
Transaction Analysis	List transactions above threshold with counterparties and amounts	Transaction edge weights, frequency by counterparty, velocity anomalies
Network Analysis	Describe transaction patterns and counterparty descriptions	Circular flows, community detection, path analysis
Contradiction Findings	Identify discrepancies between transcript statements and documentary evidence	Graph edges contradicting stated relationships
Red Flags	Identify structuring, layering, round-tripping, velocity spikes	Anomalous edge patterns, closed loops, cluster outliers
Chronology	Build event timeline across investigation date range	Temporal edge sequence across entity network
Regulatory Exposure	Identify applicable regulatory triggers in observed behaviour	Network topology consistent with known typologies
Preliminary Findings	Synthesise findings and risk indicators with evidence score	Confirmation of graph-detected patterns

13.2 Section Output Format

Each section is delivered with:

- The finding text with inline citation tags.
- Calibrated evidence score for that section.
- Citation map: every claim mapped to its chunk_id, source document, and page reference.
- Gap flags: evidence the system could not find, requiring manual investigation.
- Contradiction flags: discrepancies between transcript evidence and documentary or graph evidence.
- Iteration count: how many reasoning cycles produced the finding.

13.3 Human Review Protocol

Investigators review sections in a section-by-section interface. Each section presents the finding alongside its full citation map. The investigator can edit, annotate, and approve. Transcript-derived findings carry an additional approval flag. Approved sections lock. No section enters the final report without explicit approval.

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14. The Real Value — Institutional Intelligence

Time savings is the weakest possible framing for this architecture. It invites buyers to evaluate it against headcount reduction. The architecture is not a productivity tool — it is an institutional intelligence capability. The primary value operates on four axes that are categorically more significant than analyst hours saved, and that apply regardless of industry.

14.1 The Four Value Axes

Axis	Value Delivered	Applies To
Regulatory Compliance Velocity	Faster determination time reduces the risk of breaching mandatory filing windows. Value is penalty avoided, not hours saved.	Any regulated investigation with mandatory reporting obligations
Network Discovery	Fraud rings, related party networks, and collusion patterns that no individual investigator would detect across independently opened cases.	Financial crime, procurement fraud, corporate corruption, insurance fraud
Regulatory Defense	A complete, hash-verified audit trail with retrieval-verified findings is materially more defensible in regulatory examination or legal proceedings than a manually constructed record.	Any investigation whose outputs may face external scrutiny
Investigation Quality Floor	Every case begins with the same quality of evidence synthesis and hypothesis generation regardless of analyst experience. The floor rises. The ceiling remains with the investigator.	Any multi-investigator team with variable experience levels

14.2 Conservative Value Model by Profile

Profile	Primary Value Driver	Secondary Value Driver	Cost Anchor
1 — Minimal Compliant	Regulatory defensibility of basic investigation outputs	Consistent process quality	Infrastructure only — no AI API cost
2 — Interview-Led	Contradiction detection across interview sessions	Systematic statement management	Infrastructure + transcript processing
3 — Document Investigation	Evidence synthesis speed across large document sets	Consistent finding quality	Infrastructure + optional SOTA API
4 — Corporate Fraud	Network discovery across entity structures	Transcript-document contradiction detection	Infrastructure + local 70B model
5 — Financial Crime Full	Fraud network discovery + regulatory filing compliance	Cross-case intelligence	Infrastructure + SOTA API ($500–$1,500/month conservative)[1]
6 — Enterprise Forensic	Institutional memory compounding over case history	Court-grade evidence integrity	Full stack + fine-tuning + notarisation

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[1] Cost assumption: 200 cases/month, ~600 SOTA API calls, average context 8K tokens, 10% retry rate. Actual cost scales with case complexity and corpus size.

HONEST FRAMING

The ROI case for Profiles 1–4 does not require quantifying fraud losses or regulatory penalties. The value is defensibility and consistency — two things that are difficult to quantify and easy to lose. For Profiles 5–6, the asymmetric case is compelling: a single fraud network detected and disrupted, or a single regulatory examination navigated successfully, may return the full platform cost many times over.

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15. Deployment Constraints and Mitigation Framework

Any regulated institution deploying this architecture will encounter the following constraints. Each is addressed with a concrete mitigation. The framework is jurisdiction-agnostic — map the constraints and mitigations to your regulatory context.

C1 Data Residency and Sovereignty

> **ASSUMPTION**
> All data stores can be provisioned in a single environment.

CONSTRAINT: Most regulated jurisdictions require financial investigation data to remain within national infrastructure boundaries. Cross-border deployments require separate instances, not a shared environment.

MITIGATION: Design the stack as jurisdiction-portable from day one. Each deployment is a self-contained instance: its own Qdrant namespace hierarchy, its own graph DB, its own audit log. Only the orchestration layer can be shared. Document instance isolation before any procurement conversation.

C2 Regulatory Approval for AI in Reporting Workflows

> **ASSUMPTION**
> The system can be deployed and used immediately to support regulatory filing.

CONSTRAINT: Most financial regulators require institutions to document, and in some cases pre-approve, AI systems that touch regulatory reporting workflows. The approval process timeline is unpredictable — weeks to months depending on jurisdiction.

MITIGATION: Stage the deployment. Phases 1–5 can operate as an internal investigation tool without regulatory approval. Phase 6 (report generation) is gated behind a separate compliance and legal review. Build explainability documentation as a regulatory submission artefact from day one, not retrospectively.

C3 LLM Probabilistic Reliability

> **ASSUMPTION**
> The agent model reliably performs extraction, interpretation, and hypothesis generation.

CONSTRAINT: LLM reasoning is probabilistic. The same input can produce different outputs across runs. In an evidentiary context, this is a defensibility problem.

MITIGATION: Three controls in combination: (1) structured extraction layer with schema validation before any entity enters the graph; (2) temperature zero for all extraction and finding tasks — deterministic sampling within a run; (3) retrieval verification as primary grounding control — claims not recoverable from the corpus are stripped deterministically. Cross-run reproducibility is governed by model version pinning (B7).

C4 Entity Resolution Quality

> **ASSUMPTION**
> Entities can be reliably resolved across heterogeneous documents.

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CONSTRAINT: Real financial data is dirty. Name variants, transliterations, informal references, and data entry errors mean the same entity appears in multiple forms. Poor resolution produces a fragmented graph where circular flows are undetectable.

MITIGATION: Three-tier resolution with HITL confirmation for all non-exact matches. Resolution decisions are themselves audited investigation findings.

C5 Cross-Case Graph Governance

> **ASSUMPTION**
> Cross-case entity analysis is operationally and legally straightforward.

CONSTRAINT: Querying across cases introduces legal and governance complexity. Each case may have a different legal basis and different data subjects. Uncontrolled cross-case analysis can contaminate cases or expose the institution to legal challenge.

MITIGATION: Cross-case analysis operates on de-identified tokens only. Cross-case queries generate leads, not findings. Access restricted to senior investigators and compliance officers.

C6 Deployment Timeline in Regulated Environments

> **ASSUMPTION**
> Technical build complete in 20 weeks.

CONSTRAINT: Security review, change management, legal review of AI-generated outputs, procurement process, and stakeholder alignment routinely add 12–20 weeks in regulated institutions. The technical build is not the bottleneck.

MITIGATION: Conservative planning assumption: 32–40 weeks to production. Stage-gate the deployment so each phase delivers standalone value. Engage security, legal, and compliance teams in parallel with technical build, not sequentially.

C7 SOTA API Cost at Production Load

> **ASSUMPTION**
> SOTA API cost is $156/month at 200 cases.

CONSTRAINT: At production load including longer contexts, retries, parallel section generation, and reasoning loop iterations, realistic cost is $500–$1,500/month. Cost scales with case complexity and corpus size.

MITIGATION: The ROI case holds under all scenarios. Even at $1,500/month, SOTA API cost is under 2% of total platform cost. Monitor API cost per case from first month of production and adjust tier activation thresholds if costs exceed plan.

C8 Analyst Skill Development Under AI Assistance

> **ASSUMPTION**
> The investigation team will use the system effectively and develop alongside it.

CONSTRAINT: If junior analysts receive AI-drafted findings from day one, they may not develop the investigative reasoning skills needed to review those findings critically. Over time this produces a team that can approve AI outputs but cannot challenge them.

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MITIGATION: Two-track deployment for first 6 months: junior analysts work through the investigation framework manually first, then compare their reasoning to the AI output. The AI reasoning loop is made visible during review, not hidden behind polished output.

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11. Deployment Profiles

Six reference profiles demonstrate how the modular architecture assembles for different investigation contexts. These are starting points — not prescriptions. Every institution will adapt based on its specific regulatory obligations, evidence environment, and operational maturity.

PROFILE 1 — Minimal Compliant

Clusters active: Base only | Components: 8 | Use when: simple internal investigations, basic HR matters, low-volume case environments with manual evidence handling What this delivers: Every AI-assisted finding is human-approved, source-cited, and tamper-evidently logged. The output is defensible regardless of how simple the underlying tooling is.

PROFILE 2 — Interview-Led Investigation

Clusters active: Base + B | Components: 11 | Use when: HR investigations, whistleblower programmes, witness interview management, behavioural compliance, corruption investigations where testimony is the primary evidence type What this delivers: Speaker-turn evidence retrieval, contradiction detection across interview sessions, separate approval gate for statement-derived findings, complete audit trail of what was said, when, by whom, and what finding it supported.

PROFILE 3 — Document Investigation

Clusters active: Base + A + D | Components: 20 | Use when: legal due diligence, contract dispute, procurement review, audit investigation, regulatory examination preparation What this delivers: Full document ingestion and retrieval across any file type, recursive hypothesis formation across large document sets, calibrated evidence scoring, parallel report section generation.

PROFILE 4 — Corporate Fraud & FCPA/ABAC

Clusters active: Base + A + B + C + D | Components: 28 | Use when: bribery investigation, conflict of interest mapping, third-party due diligence, beneficial ownership analysis, multi-party corporate fraud What this delivers: Full documentary and transcript evidence processing, entity network mapping across corporate structures, contradiction detection between interview statements and documentary evidence.

PROFILE 5 — Financial Crime Full Deployment

Clusters active: Base + A + B + C + D + E | Components: 31 | Use when: AML, fraud investigation, transaction monitoring follow-up, STR generation — high-volume, multi-entity, external AI required What this delivers: Everything in Profile 4 plus privacy-preserving access to frontier AI reasoning via pseudonymisation gateway. Transaction network detection, circular flow analysis, cross-case fraud ring discovery.

PROFILE 6 — Enterprise Forensic Full Stack

Clusters active: Base + A + B + C + D + E + F + G | Components: 35 | Use when: serious fraud investigation, enforcement agency, central bank examination, Big 4 forensic practice What this delivers: Full stack including institutional memory through historical case fine-tuning and court-grade evidence integrity through external notarisation. The system improves with every case.

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11.1 Profile Component Matrix

Component Group	Profile 1	Profile 2	Profile 3	Profile 4	Profile 5	Profile 6
Mandatory Base (B1–B8)	✓	✓	✓	✓	✓	✓
Cluster A — Documentary	—	—	✓	✓	✓	✓
Cluster B — Behavioural	—	✓	—	✓	✓	✓
Cluster C — Network Intelligence	—	—	—	✓	✓	✓
Cluster D — Advanced Reasoning	—	—	✓	✓	✓	✓
Cluster E — Privacy Gateway	—	—	—	—	✓	✓
Cluster F — Institutional Memory	—	—	—	—	—	✓
Cluster G — Integrity Elevation	—	—	—	—	—	✓
Total Components	8	11	20	28	31	35

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12. Worked Case Walkthrough — Profile 4 Deployment

This walkthrough demonstrates a Profile 4 deployment — Base + Clusters A, B, C, D — using a fictional corporate fraud investigation. All entities, individuals, and amounts are illustrative only. The scenario: a mid-size manufacturing company suspects procurement fraud involving a senior operations manager and a network of related suppliers.

PROFILE IN USE

Base + Cluster A (documents) + Cluster B (interview transcripts) + Cluster C (entity network) + Cluster D (advanced reasoning). Cluster E not active — full local model stack, no external API. Cluster G not active — internal investigation, not yet at proceedings stage.

Step 1 — Case Open and Evidence Upload

- Investigator opens CASE_2026_0203. Uploads: 18 months purchase orders (CSV), 3 supplier contracts (PDF), expense reports (Excel), 6 email chains (MSG), 2 interview transcripts (DOCX).
- Cluster A activates: ingestion pipeline processes all files. 1,240 chunks indexed to CASE_2026_0203_ns.
- Cluster B activates: 2 transcript files chunked at speaker-turn level. 84 speaker turns indexed with interviewer ID and timestamps.
- Audit log: 7 ingestion events recorded with file hashes. Hash chain initialised.

Step 2 — Entity Resolution (Cluster C)

- Entity extractor identifies 31 entities: 4 individuals, 12 suppliers, 8 internal cost centres, 7 bank accounts.
- Tier 1 auto-resolve: 3 supplier entities matched across PO CSV and contract PDFs on registration number. Logged. No investigator action.
- Tier 2 blocking: 'Apex Solutions Ltd' in contracts matches 'Apex Solns' in PO data — 81% confidence. Investigator confirms merge.
- Tier 3 blocking: Director of Apex Solutions shares a residential address with the subject manager. System flags as possible alias. Investigator marks as 'confirmed related party' — linked with relationship edge, not merged.
- Graph builds: 28 resolved nodes. Related party edge flagged and visible in graph.

Step 3 — Graph Analysis (Cluster C)

- Graph query: counterparty concentration. Result: 67% of subject manager's approved POs flow to 3 suppliers — Apex Solutions, Pinnacle Supplies, and TradeCo. All three registered within 8 months of each other.
- Graph query: shared attributes. Result: Apex Solutions and Pinnacle Supplies share a registered agent. TradeCo director's phone number appears in subject manager's personal expense claims.
- Cross-case index query: TradeCo token appears in one prior closed case — flagged as unresolved lead 14 months ago. Senior investigator notified.

Step 4 — Recursive Reasoning (Cluster D)

- Iteration 1: Planner queries PO approval patterns. Analyzer: subject manager sole approver on 89% of flagged supplier POs. Framework coverage 34%. Evidence score: 0.38.

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- Iteration 2: Planner queries supplier registration timeline vs manager's tenure. Analyzer: all 3 suppliers registered after manager joined. Invoice amounts consistently just below single-approval threshold. Score: 0.61.
- Iteration 3: Planner queries transcript — manager's statement on supplier selection. Cluster B: manager states [chunk TRN_014] 'these suppliers were already on the approved list when I arrived.' Contradiction Detector: approved vendor list at date of joining does not include any of the 3 suppliers. Score: 0.84.
- Iteration 4: Planner queries expense claims linked to TradeCo director's contact. Retrieval Verifier confirms. Evidence score: 0.91. All sections lock.

Step 5 — Report Generation and Human Review

- 8 report sections generated in parallel. Total time: 81 seconds.
- Average evidence score across sections: 0.88. Citation map: 52 distinct source citations. 0 claims stripped by Retrieval Verifier. 3 gaps flagged for manual follow-up.
- Investigator reviews section by section. Edits the preliminary findings section to add context on the prior TradeCo lead. Approves all sections.
- Final report locked. Audit chain: 187 immutable log entries from case open to report lock.

OUTCOME

Investigation open to draft report: 6 hours. Three related-party suppliers identified through graph — none visible from document review alone. Transcript contradiction between manager's statement and vendor list is a material finding, detected automatically. Prior case link on TradeCo generates active lead. Every finding source-cited, every AI action audited.

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15. Competitive Positioning

The investigation technology landscape spans five categories. Each addresses part of what regulated investigation requires. None encodes investigation methodology as engineering constraints across the full spectrum from evidence grounding to audit immutability.

Category	Representative Tools	What They Deliver	What Is Missing vs This Architecture
Financial Crime Platforms	Palantir Gotham, Chainalysis, Feedzai, NICE Actimize	Transaction monitoring, entity network visualisation, alert generation, risk scoring	Investigation methodology encoding, recursive hypothesis formation, claim-level evidence grounding, HITL approval chain, cross-cluster modularity for non-financial contexts
Legal Document Review	Relativity, Reveal, Everlaw, Nuix	Large-scale document search and review, privilege review, keyword and concept search	Graph intelligence, entity resolution with HITL, recursive reasoning, audit immutability to investigation standard, report generation with evidence grounding
HR & Compliance Investigation	i-Sight, NAVEX Global, Convercent, Case IQ	Case management, intake workflow, reporting dashboards, basic document attachment	AI-assisted evidence synthesis, contradiction detection across interview transcripts, entity network mapping, calibrated evidence scoring, hash-chain audit
Audit Management	AuditBoard, TeamMate+, Diligent	Control testing, audit programme management, finding workflow, remediation tracking	Investigation-depth reasoning, hypothesis formation, transcript intelligence, cross-case intelligence, evidence grounding at claim level
Generic RAG / LLM Tools	LangChain deployments, custom GPT wrappers, general vector search	Document retrieval, conversational query, basic summarisation	Case isolation by architecture, entity resolution, graph intelligence, retrieval verification, methodology encoding, human approval gates, audit immutability

THE DIFFERENTIATOR

Every tool in the table above applies AI to investigation work. This architecture encodes investigation methodology as engineering constraints — and then applies AI within those constraints. The difference is not capability. It is accountability. Every other tool produces outputs that an investigator works with. This architecture produces outputs that an investigator is accountable for, because every output required their explicit approval before it entered the record.

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16. Conclusion

The modular architecture described in this paper is not a product for financial institutions. It is a framework for any organisation that conducts regulated investigations and needs its outputs to be defensible, auditable, and traceable to evidence.

The eight mandatory base components are non-negotiable regardless of industry, jurisdiction, or AI maturity. They encode the minimum standard for a defensible investigation output: case isolation, evidence traceability, human accountability, and tamper-evident audit. Above the base, seven modular clusters assemble to match the investigation context — from a two-cluster interview-only deployment to a full seven-cluster enterprise forensic platform.

The deployment profiles demonstrate that this is not a binary choice between 'basic tool' and 'full platform.' There is a continuous upgrade path. An organisation that starts with Profile 2 for its interview programme can add Cluster A when it begins processing documents, then Cluster C when its cases become multi-entity, then Cluster D when volume and complexity justify recursive reasoning. Each addition builds on the base without replacing it.

The architecture's defensible moat is not technical. Any sufficiently resourced team can implement the components described here. The moat is methodological: the investigation framework encodes what the system must do, and the technology serves that framework. That inversion — methodology first, technology second — is what distinguishes an Institutional Reasoning Engine from a sophisticated document search tool.

> **CORE PRINCIPLE**
>
> Methodology shapes technology. The investigation framework determines what the system is built to do. The technology is in service of the methodology — never the other way around.

— End of White Paper —

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Appendix A — Real-World Operational Case

This appendix describes how an institution actually deploys and operates this architecture — not how a case runs through the system (Section 12), but how the system is built, stood up, and operated over 18 months in a real organisational environment.

A.1 — The Decision to Deploy

Trigger	Institution Type	Immediate Driver
Regulatory examination finding	Bank, insurer, payments firm	Examiner notes that investigation files lack consistent evidence trails.
Missed reporting window	AML/fraud team	Manual investigation process caused a statutory filing to be submitted 4 days late.
Senior leadership mandate	Corporate compliance function	Board-level directive following an external fraud incident: 'Show us the investigation process is systematic and auditable.'
Capacity crisis	Forensic practice or enforcement unit	Case backlog has grown to 6 months. Existing analyst headcount cannot close cases at intake rate.

Before deployment begins, three prerequisites must be satisfied: (1) the investigation methodology must be documented; (2) data inventory must be completed; (3) legal and compliance sign-off on AI in investigation workflows.

A.2 — Selecting the Deployment Profile

INSTITUTION 1 — Mid-size cooperative bank, 40 investigations/year

Evidence: transaction CSVs, KYC PDFs, some interview transcripts. Cases are mostly single-subject. Profile selected: Profile 3 (Base + A + D). Rationale: documentary evidence is the primary input; cases are not complex enough to justify graph intelligence; 40 cases/year means 5-year timeline to reach fine-tuning threshold.

INSTITUTION 2 — Corporate ethics and compliance team, 120 investigations/year

Evidence: primarily interview transcripts and emails. Some cases involve related parties and vendor networks. Profile selected: Profile 2 initially, upgrading to Profile 4 at month 6. Rationale: interview transcripts are the immediate pain point — phased deployment delivers early value.

INSTITUTION 3 — National fraud enforcement unit, 500+ investigations/year

Evidence: all types. Complex multi-party cases. Outputs go to courts and regulators. Profile selected: Profile 6 (Full Stack). Build sequence: mandatory base first, Clusters A and B (months 1–3), C and D (months 4–6), E (months 6–8), G (months 8–10), F after 200 cases close.

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A.3 — The Build Sequence

Phase	Duration	What Is Built	Exit Criterion
Phase 1 — Base	Weeks 1–6	Case isolation tested with synthetic cases. Audit logger verified. Hash chain running. Human review gate operational. Model version pinning confirmed.	10 synthetic cases run end-to-end. Audit chains verified. Zero cross-case contamination. Human review gate blocks record without approval.
Phase 2 — First Cluster	Weeks 7–14	First activated cluster built and tested. For most institutions: Cluster A.	50 real documents ingested and retrieved accurately. Retrieval verifier confirmed stripping unverifiable claims.
Phase 3 — Parallel Run	Months 4–9	Investigators work both old and new processes on the same cases. New system outputs reviewed but not used as primary record.	Investigation quality metrics baseline established. Edge cases identified and resolved.
Phase 4 — Primary System	Month 10 onwards	New system becomes primary investigation record. Old process retired.	Zero critical defects in parallel run period. Regulatory and legal sign-off obtained.

A.4 — First 90 Days of Operation

- Ingestion errors on edge-case formats: scanned PDFs without OCR, password-protected files, non-standard CSV encodings. Resolution: build a document preprocessing checklist.
- Entity resolution queue growth: Tier 2 and Tier 3 decisions accumulate faster than investigators clear them. Resolution: one entity resolution review session per week, calendar-blocked.
- Heavy human editing of AI findings: in the first 30 days, investigators typically edit 60–70% of AI-generated draft findings materially. This is not a system failure — it is the system working correctly.
- Investigators approving without reading: governance response: periodic audit of approval patterns. Any investigator approving more than 95% of findings without edits triggers a quality review.
- First cross-case match: typically occurs within 60 days. Ensure the lead review process is operational before the first case closes.

A.5 — Steady-State Operations

Metric	Baseline (Month 1)	Mature (Month 18)	What It Indicates
Average evidence score at report lock	Not measured — manual process	0.85–0.91	Consistent evidence quality across cases and investigators
% findings requiring material human edit	60–70%	20–35%	Retrieval templates refined to match institutional methodology
Investigation open-to-draft time	5–8 days (manual)	4–8 hours (AI-assisted)	Analyst time freed for judgment work
Cross-case leads generated	0 — cases siloed	3–8 per quarter	Network visibility that was structurally impossible before

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Metric	Baseline (Month 1)	Mature (Month 18)	What It Indicates
Audit chain requests from regulators	Rare	Handled in minutes	Institution can respond with mathematical proof, not assertion

A.6 — What Can Go Wrong

> **HONEST ASSESSMENT**
> Three failure modes that recur across deployments. Not theoretical — these are the operational risks that undermine even well-designed systems.

- The base is bypassed. Investigators find workarounds to skip the human review gate. Mitigation: the gate is the record, not a step before the record. Make this architecturally true, not policy-dependent.
- Entity resolution queue blocks investigation. When Tier 2 and Tier 3 decisions accumulate unreviewed, Cluster C becomes a bottleneck. Mitigation: dedicated review sessions, SLA on Tier 2/3 clearance (48 hours maximum).
- Fine-tuning applied too early. A model fine-tuned on 80 cases overfits to a non-representative sample. Mitigation: enforce the 200-case minimum without exceptions.

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Appendix B — Investigation Typology Reference Matrix

Use this matrix to find your investigation context, identify the appropriate deployment profile, and locate the relevant sections of this paper.

Investigation Type	Profile	Clusters	Primary Evidence	The Module That Makes the Difference	Illustrative Finding
AML / Transaction Layering	5	A+C+D+E	Transactions, KYC, counterparty records	Graph — circular flow detection	3-hop circular flow, 18 days, 3 shell entities registered same agent
Procurement Fraud	4	A+B+C+D	POs, contracts, vendor records, interviews	Contradiction Detector	Vendor registered 3 months after manager joined; manager states vendor was pre-existing
Insurance Fraud Ring	5	A+C+D	Claims, witness statements, repair records	Cross-case entity index	6 independently filed claims share 2 witnesses and 1 contractor across 3 adjusters
FCPA / Bribery	4	A+B+C+D	Emails, expense claims, contracts, interviews	Entity Resolution + Contradiction Detector	Consultant and undisclosed agent resolved as same individual across 3 document types
HR Misconduct	2	B only	Interview transcripts, communications	Transcript Contradiction Detector	Statement in session 3 contradicts statement in session 1 on key timeline fact
Insider Trading	3	A+D	Trade records, communications, calendar data	Recursive Reasoning Loop	Hypothesis iterates from coincidence to pattern across 7 trading events
Regulatory Examination Prep	3	A+D	Policy documents, audit reports, board minutes	Recursive Reasoning + Evidence Scorer	Board resolution contradicting current policy identified in iteration 3
Beneficial Ownership Fraud	4	A+C+D	Corporate filings, KYC, bank records	Graph Query Layer	4-layer ownership chain resolves to same ultimate beneficial owner
Whistleblower Investigation	2–3	B+A	Transcripts, corroborating documents	HITL Approval Gate + Transcript Gate	Statement-derived finding requires separate approval before entering record
Healthcare Billing Fraud	5	A+C+D+E	Claims, patient records, provider records	Cross-case entity index	Single provider appearing in 23 independently opened claims
Trade-Based Money Laundering	6	A+C+D+E+F	Trade documents, invoices, shipping records	Institutional Memory (Cluster F)	Fine-tuned model recognises over-invoicing pattern — 40%+ margin spread typology
Cybercrime / Digital Fraud	4	A+C+D	System logs, communications, access records	Graph + Recursive Reasoning	IP address entity resolved across 8 independently opened cases

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HOW TO USE THIS MATRIX

Find your investigation type. Note the profile. Go to Section 11 for the deployment profile detail and Section 2.3 for the dependency map. If your investigation type is not listed, identify the closest match by evidence type — the evidence environment determines the profile more than the investigation label does.

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Appendix C — Plain-English Glossary

Technical terms used in this paper, explained for non-technical readers.

Term	Plain English
Institutional Reasoning Engine (IRE)	An AI system designed for regulated investigations. It forms a hypothesis, iterates through evidence, requires human approval at every finding, and produces a tamper-evident audit record.
RAG (Retrieval-Augmented Generation)	A technique where the AI only reasons about documents you have provided, not its general training data. Every claim traces back to a specific document.
HITL (Human-in-the-Loop)	A design requirement that a human must review and approve an AI output before it becomes an official record.
Hash Chain	A cryptographic technique where each audit log entry is mathematically linked to the previous one. Any tampering breaks all subsequent links — making it detectable.
Namespace Isolation	Each case gets its own sealed evidence container. The AI cannot accidentally mix evidence from different cases — by design, not policy.
Temperature = 0	A setting that makes an AI model produce the same output for the same input within a single run. Used for all extraction and finding tasks.
SOTA API	State-of-the-art external AI model (e.g. GPT-4, Claude). Used only for final risk narrative generation; never receives real personal data.
Pseudonymisation	Replacing real names and account numbers with anonymous tokens before sending data to any external system.
Fine-Tuning	Training an AI model on the institution's own completed investigation cases so it learns your specific fraud patterns.
Cypher	The query language used to ask questions of graph databases (like Neo4j). Equivalent to SQL for relationship-based data.
Namespace	A labelled storage partition. Think of it as a sealed folder — data in Case_001_ns cannot be accessed when working on Case_002_ns.
Vector Store	A database that stores documents as mathematical representations (vectors) so they can be searched by meaning, not just keyword.
STR / HLIR / SLIR	Suspicious Transaction Report / High Level Investigation Report / Summary Level Investigation Report — standard output formats for regulated financial investigations.
Embedding Model	An AI model that converts text into a mathematical representation for semantic search. Runs locally — no data leaves the institution.
Schema Validation	A check that AI-extracted data matches the required format before it enters any system. Like a form that rejects invalid entries.
Inference Re-identification	The risk that an AI model, given pseudonymised data, produces output that indirectly reveals who the data subject is.
QLDB	Amazon Quantum Ledger Database — an immutable, cryptographically verifiable ledger used to anchor audit chain hashes externally.

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Appendix D — What This Is / What This Is Not

A single-page reference for any reader who wants to quickly orient before reading the full paper.

This Architecture IS	This Architecture IS NOT
A methodology encoding framework — investigation process as engineering constraints, not advisory policies.	A replacement for investigator judgment — every material output requires human approval.
A modular system scaling from 8 base components to a 35-component enterprise forensic platform.	A product available for purchase — this is an architectural blueprint.
Jurisdiction-agnostic — applicable to any regulated investigation in any industry.	A compliance guarantee in any specific jurisdiction.
An AI acceleration layer with mandatory human gates at every material finding.	An autonomous AI investigation system.
Tamper-evident by design at the base level; court-grade with Cluster G active.	Unconditionally immutable in all configurations.
Open to any deployment scale — from a 2-person HR team to a national enforcement unit.	Dependent on any specific AI vendor or cloud provider.
Upgradeable incrementally — start at Profile 1, grow to Profile 6 without rebuilding.	A system that requires full deployment before delivering any value.

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